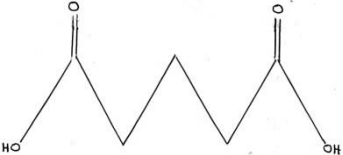


Question				Marking details	Marks available					
					AO1	AO2	AO3	Maths	Prac	Total
	(c)	(i)		award (1) for any of following hydrochloric acid (in the presence of zinc chloride) phosphorus pentachloride phosphorus trichloride thionyl chloride accept correct formulae		1			1	1
		(ii)		ammonia accept NH ₃		1				1
		(iii)		nucleophilic substitution	1					1
		(iv)					1			1
	(d)			M_r benzene-1,4-dicarboxylic acid is 166 (1) M_r 'repeating unit' is 192 (1) mass of acid = $75 \times \frac{166}{192} \times \frac{90}{100} = 58.4 \text{ kg}$ (1)		1 1 1		1		3